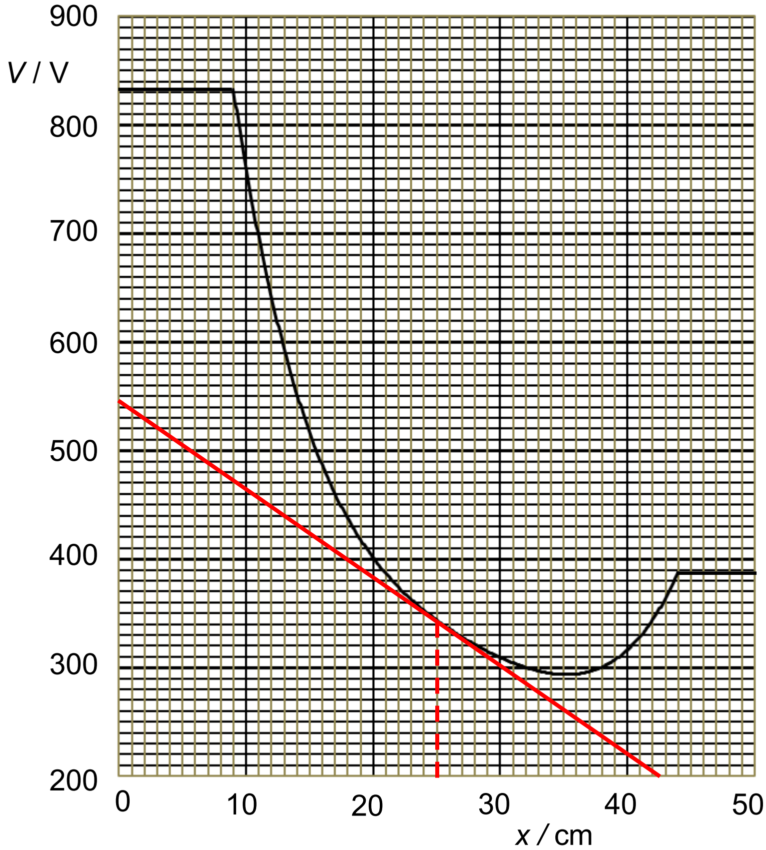
Q5		
(a)(i)	Direction of the electric field at x = 25.0 cm is towards the right. Electric field is always directed towards lower potential. OR Direction of the electric field at x = 25.0 cm is towards the right. The potential gradient is negative at this value of x as can be observed by drawing a tangent to the curve at x = 25.0 cm. Since E is (the negative) of the potential gradient, E takes on a positive value, which means the electric field is directed towards the right.	
(a)(ii)	 Draw a tangent to the curve at x = 25.0 cm Potential gradient $\frac{dV}{dx} = \frac{545-200}{0.0-42.5} = -8.12$ (3 s.f.) Electric field $E = -\frac{dV}{dx} = 8.12\text{ V cm}^{-1}$ (acceptable range: 7.3 – 9.7)	
(a)(iii)	Mobile charge carriers within a conductor will always re-distribute until a certain equilibrium state where **the electric field within it is zero**. Zero electric field means that there is **zero potential gradient**, and hence **constant potential.**	
(b)	The ion will accelerate to the right until ***x* = 35.0 cm**, then the ion will decelerate and momentarily come to rest at ***x* = 42.0 cm**, and accelerate back to the left and momentarily stops at **x = 25.0 cm** before accelerating to the right again, **repeating** the motion.	

	OR The ion will oscillate between the points ***x* = 25.0 cm and *x* = 42.0 cm**.	